

# NeuroScan AI v3.0 — Clinical Imaging Report

AI-assisted neuro-radiological screening (research use only)

| Report ID    | NS3-20260913-093731               | Generated    | 13 Sep 2026 09:37 |
|--------------|-----------------------------------|--------------|-------------------|
| Source       | brisc2025_test_00701_pi_ax_t1.jpg | Input domain | full_image        |
| Deployed arm | full_image                        | Backbone     | EfficientNet-B0   |

## PRIMARY AI IMPRESSION

| AI impression          | PITUITARY                        |
|------------------------|----------------------------------|
| Ensemble confidence    | 92.29%                           |
| Risk stratum           | MEDIUM                           |
| Predictive uncertainty | sigma = 0.0158 (Low)             |
| Branch consensus       | 100% of the three branches agree |

## CLASS POSTERIOR

| Class      | Ensemble | Standard | MC dropout | TTA    | MC sigma |          |
|------------|----------|----------|------------|--------|----------|----------|
| glioma     | 0.82%    | 0.61%    | 0.65%      | 1.33%  | 0.0050   |          |
| meningioma | 6.04%    | 2.59%    | 2.75%      | 15.92% | 0.0222   |          |
| no_tumor   | 0.85%    | 0.70%    | 0.84%      | 0.87%  | 0.0072   |          |
| pituitary  | 92.29%   | 96.11%   | 95.76%     | 81.88% | 0.0289   | SELECTED |

Fusion weights 0.00 / 0.75 / 0.25 (standard / MC dropout / TTA) were selected by grid search on the validation split; the held-out test split took no part in choosing them.

## LESION GEOMETRY (Attention U-Net)

| Area     | 414 px <sup>2</sup> (0.96% of the crop) | Circularity | 0.472       |
|----------|-----------------------------------------|-------------|-------------|
| Centroid | (114, 90)                               | Quadrant    | Upper-Right |

## INTERPRETATION

Imaging features are consistent with a sellar or parasellar lesion. Endocrine evaluation for hormonal hypersecretion and visual field assessment for chiasmal compression are indicated.

**Recommendation:** Endocrinology and neurosurgery referral; pituitary hormone panel and visual fields.

## MODEL PROVENANCE (measured, not asserted)

| Dataset             | BRISC2025 (briscdataset/brisc2025) — 4211 train / 744 val / 995 held-out test                         |
|---------------------|-------------------------------------------------------------------------------------------------------|
| Test split          | Official BRISC2025 test folder, untouched during training                                             |
| Segmentation        | Attention U-Net on annotated masks — test Dice 0.8400, IoU 0.7756                                     |
| Deployed classifier | full_image — test accuracy 98.59%, macro F1 98.62%                                                    |
| With ensemble       | test accuracy 98.79%, macro F1 98.84%, kappa 0.9835                                                   |
| Controlled contrast | full-image arm macro F1 98.62% vs ROI arm 97.56%; McNemar p = 0.0433                                  |
| Explainability      | Grad-CAM on the last backbone convolution, computed on the same input domain the model was trained on |
| Uncertainty         | MC dropout, 30 stochastic passes (BatchNorm frozen)                                                   |

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*IMPORTANT: this is an experimental research system. This report is not a medical diagnosis, has not been reviewed by a licensed clinician, and must not be used to direct patient care. Model outputs are derived from a single 2D slice and must be correlated with the full study, clinical history and expert radiological interpretation. The authors accept no liability for decisions based on this output.*